Shifts in Processing Inclusive Language in Argentine Spanish (2019 & 2025)

Ana Zarwanitzer, Ignacio Spiousas, Santiago Estremero, Gala Esperanza Coronas, Carlos Gelormini-Lezama (cgelormini@udesa.edu.ar)

Introduction: Over the past decade, inclusive morphemes have emerged in Argentina and other Spanish-Speaking countries—a phenomenon also observed in other languages (Penfield, 1987; MacKay & Fulkerson, 1979; Vergoossen, 2019; Arnold, Mayo, & Dong, 2020, 2021; Zunino & Tomé Cornejo, 2025). This study replicates a self-paced sentence-by-sentence reading experiment from 2019 in River Plate Spanish, which showed that sentences with inclusive language yielded longer reading times than generic masculine (Zarwanitzer & Gelormini-Lezama, 2023). Two interpretations were proposed at the time: (1) speakers reject inclusive language as unacceptable and (2) the increased reading times only reflects lower exposure.

Method: A set of 18 items like the one in Table 1 was constructed. Sentence 1 introduced a context of place/time. Sentence 2 contained an animate noun like *hijos* ("children") in three conditions: generic masculine morpheme *-o*, and two versions of inclusive language, *-e* and *-x*. Participants included 85 monolingual speakers of River Plate Spanish tested in 2019 (39 women, 46 men; age range 18–71, M = 38.3, SD = 16.2) and 39 monolingual speakers of River Plate Spanish tested in 2025 (30 women, 9 men; age range 19–58, M = 28.2, SD = 11.2). The dependent variable was the reading time of Sentence 2. Trials exceeding each participant's median ± 3 MAD were excluded (~5%).

Results: Table 2 shows the mean reading times of Sentence 2. We fit a linear-mixed model with sentence condition (*-o*, *-e*, *-x*), experiment year (2019, 2025), and their interaction as fixed effects, and with crossed random intercepts for subjects and items. The model showed a significant interaction, along with main effects of sentence condition, and experiment year. To further explore the interaction, we calculated simple contrasts comparing the levels of condition at each level of year (Tukey corrected). For both years, reading times were longer for the *-e* condition than for *-o* (2019: 464.4 ms., $p < 0.001$; 2025: 286.6 ms., $p < 0.001$). However, the comparison between *-e* and *-x* changed over time: in 2025, *-e* elicited significantly longer reading times than *-x* (203.0 ms, $p < .02$), whereas in 2019 the difference was not significant (65.9 ms, $p = .40$). Additionally, *-x* yielded longer reading times than *-o* in 2019 (398.6 ms, $p < .001$), but not in 2025 (83.6 ms, $p = .48$) (see Figure 1).

A plausible explanation for the change in the processing of the *-x* morpheme may simply be due to increased exposure over time. However, the main effect of the experiment year in the model indicates that participants tested in 2025 were overall faster (79.2 ms, $p < .001$), a pattern that cannot be attributed solely to the passage of time. When we include participant age instead of experiment year as a fixed effect, we observe not only a general increase in reading times with age (17.9 ms per year, $p < .001$) but also a significant interaction: each additional year of age contributed an extra 10.4 ms for the *-e* condition and an extra 13.1 ms for the *-x* condition. This finding highlights the differential impact of age on reading times, with a general effect that is amplified for the inclusive forms (see Figure 2).

Discussion: This study shows that, from a processing standpoint, it is inappropriate to treat "inclusive language" as a unified category. A key finding is that the inclusive morpheme *-e* continues to incur processing costs, even among young readers who show no processing differences between *-x* and *-o*. One possible explanation is that *-e* may introduce ambiguity, as it can be confounded with other phonotactically well-formed Spanish lexical items. In contrast, *-x* clearly violates these constraints, making it more readily identifiable as an inclusive marker. Overall, the preference for *-x* indicates a shift in its acceptability among native speakers. This, in turn, suggests that the social and linguistic changes motivating the emergence of this morpheme provide a stable basis for its gradual consolidation into native speakers' mental grammars.

Table 1. Sample experimental item in all three conditions.

Sentence 1	Estuvo lloviendo toda la mañana. (*It was raining all morning.)*		
	Generic Masculine "o"	**Inclusive Morpheme "e"**	**Inclusive Morpheme "x"**
Sentence 2	Mis **hijos** prefieren quedarse en casa. (*My children prefer to stay at home.)*	Mis **hijes** prefieren quedarse en casa. (*My children prefer to stay at home.)*	Mis **hijxs** prefieren quedarse en casa. (*My children prefer to stay at home.)*

Table 2. Mean reading times in milliseconds and standard errors for sentence 2 in all three conditions.

	Conditions					
	Generic Masculine "o"		**Inclusive Morpheme "e"**		**Inclusive Morpheme "x"**	
Year	2019	2025	2019	2025	2019	2025
	2095 (44)	1639 (53)	2597 (57)	1958 (66)	2573 (59)	1712 (50)

Figure 1. Mean reading times per condition and year. The bars represent the global mean per condition and year. Each dot and line represents the mean and standard error of a participant.

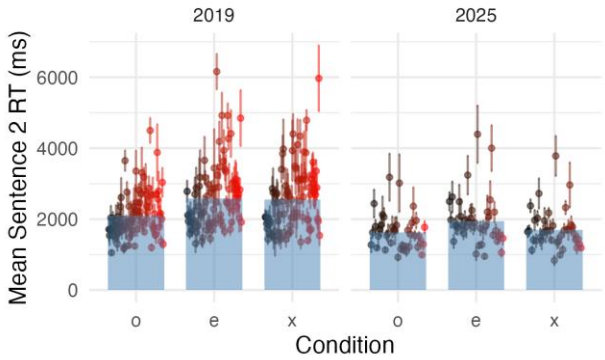
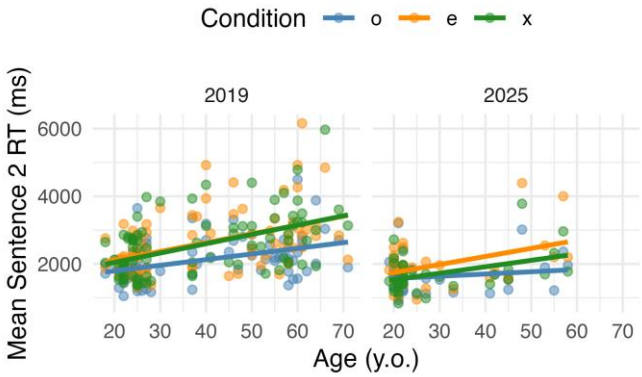


Figure 2. Relation between mean reading times and age of participant for each year. Each dot represents the mean of each subject for a given condition and the lines represent the best fitted line in the plot (not the predictions of the models).



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